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1. Overview

P2ROUD = Preclinical PROspective Undersampling in multiple Dimensions.

P2ROUD is an application with a graphical user interface to reconstruct MRI data acquired with an MR Solutions preclinical MRI system, in particular data with k-space undersampling in one or more dimensions. Besides Cartesian 2D and 3D data it reconstructs 2D radial, 3D UTE and ZTE scans, and 2D EPI scans for diffusion and fMRI, which it can export as a BIDS dataset.

P2ROUD is a stand-alone application: everything it uses is inside it, and nothing else has to be installed.

2. Contact

P2ROUD and this manual are written by:

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3. Disclaimer

This is a beta version. The software has been extensively tested using mouse and rat data acquired with an MR Solutions preclinical 7.0T/24cm system. However, this does not warrant that the functions contained in the program will meet your requirements or that the operation of the program will be uninterrupted or error-free. In case of questions or issues, please contact Gustav Strijkers or Bram Coolen.

4. Installation

macOS

Requirements: a Mac with Apple silicon (M1 or later), macOS 26 or later.

P2ROUD comes as a disk image, `P2ROUD-0.5.9-macOS-arm64.dmg`. Nothing else has to be installed: Python, dcm2niix and everything else the app uses are inside it.

1. Open the disk image and drag **P2ROUD** onto the Applications folder in its window.
2. Open it from Applications. The app is not signed with an Apple Developer ID, so the first time macOS refuses it ("Apple could not verify ..."); click **Done**. Then open **System Settings > Privacy & Security**, scroll down to the line saying P2ROUD was blocked, click **Open Anyway** and confirm. From then on it opens as any other app.

   Instead of step 2, the same in Terminal, once:

   ```
   xattr -dr com.apple.quarantine /Applications/P2ROUD.app
   ```

The first start can take a little longer: macOS checks the app once.

Windows

Requirements: a PC with Windows 10 or 11, 64-bit.

P2ROUD comes as an installer, `P2ROUD-0.5.9-setup.exe`. Nothing else has to be installed: Python, dcm2niix and everything else the app uses are inside it.

1. Run the installer. It is not signed, so the first time Windows may stop it ("Windows protected your PC"): click **More info** and then **Run anyway**.
2. The installer puts the app in `%LOCALAPPDATA%\Programs\P2ROUD` for the current user, without administrator rights (installing it for all users is an option), with a Start-menu entry and, if you choose, a desktop shortcut. At the end it can start the app.

To remove the app, use **Uninstall P2ROUD** in the Start menu, or **Settings > Apps**.

BART (optional)

With BART, the Berkeley Advanced Reconstruction Toolbox (https://mrirecon.github.io/bart/), the compressed sensing reconstructions run with BART. Without it, the app reconstructs with its own engine (sigpy), with the same settings; it takes two to six times as long. P2ROUD is tested with BART 1.0.00.

The app chooses the engine for you: BART where it is installed and can do the settings chosen, being the faster, and its own engine where BART is not installed or cannot do them (a temporal basis with **TVtime** or **LLRxyz**, see below). Each reconstruction says in the message box which engine ran and why, for example "Engine: sigpy (BART cannot do TVtime with a temporal basis) ...".

At its start the app writes BART's version in the message box, followed by "BART toolbox detected ..." or "WARNING: BART toolbox not available ...", and whether BART was compiled with GPU support. A BART built for the GPU is used on the GPU.

Installing BART on macOS with MacPorts

1. Install the Xcode command line tools:

   ```
   xcode-select --install
   ```

2. Install MacPorts from https://www.macports.org/.
3. Install the packages:

   ```
   sudo port install fftw-3-single gcc14 libpng openblas flock gmake coreutils l
   ibblocksruntime
   ```

4. Download and unpack BART:

   ```
   wget https://github.com/mrirecon/bart/archive/v1.0.00.tar.gz
   tar xvfz v1.0.00.tar.gz
   cd bart-1.0.00
   ```

5. Build and install:

   ```
   export PATH=/opt/local/libexec/gnubin:$PATH
   CC=gcc-mp-14 gmake
   sudo gmake install
   ```

   The compiler name must match the gcc version installed in step 3: gcc14 gives gcc-mp-14, gcc15 gives gcc-mp-15.

6. Check that it works: `bart version`

Installing BART on macOS with Homebrew

1. Install the Xcode command line tools: `xcode-select --install`
2. Install Homebrew:

   ```
   /bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/instal
   l/HEAD/install.sh)"
   ```

3. Install the packages BART needs:

   ```
   brew install gcc libpng fftw openblas make llvm libomp
   ```

4. Download and unpack BART as above (step 4 of MacPorts).
5. Build and install:

   ```
   CC=gcc-15 MACPORTS=0 gmake
   sudo gmake install
   ```

   The compiler name must match the version Homebrew installed; check it with `ls /opt/homebrew/bin/gcc-*`.

6. Check that it works: `bart version`

How P2ROUD finds BART on macOS

In the normal case there is nothing to configure: `sudo make install` puts BART in `/usr/local/bin`, which is where the app looks first. It looks in these folders, in this order:

Folder	
`/usr/local/bin`	where BART's own `make install` puts it
`/usr/bin`	
`/opt/homebrew/bin`	Homebrew's
`/opt/local/bin`	MacPorts'

An app opened from the Finder or the Dock does not see the settings of your shell (such as `TOOLBOX_PATH` or `PATH` in your shell profile). If your `bart` is in another folder, a link to it in `/usr/local/bin` lets the app find it. In Terminal, once, with the folder your bart is in instead of `/path/to/bart` (Terminal asks for your password):

```
sudo mkdir -p /usr/local/bin
sudo ln -s /path/to/bart/bart /usr/local/bin/bart
```

Installing BART on Windows 11

On Windows, BART runs inside the Windows Subsystem for Linux (WSL), in the default distribution; the app calls it as `wsl bart`.

1. Install WSL. Start a Windows PowerShell as administrator and run:

   ```
   wsl --install
   ```

   This enables the feature, installs WSL 2 and installs Ubuntu in one step. A restart is needed. The first time you start the Linux prompt you are asked to create a user account. On older systems, or if `wsl --install` is not recognized, see https://pureinfotech.com/install-wsl-windows-11/.

2. Update the Linux distribution (at the Linux prompt):

   ```
   sudo apt-get update
   sudo apt-get dist-upgrade
   ```

3. Install the packages BART needs (gcc 12 or newer):

   ```
   sudo apt-get install make gcc libfftw3-dev liblapacke-dev libpng-dev libopenb
   las-dev gfortran
   ```

4. Download and unpack BART:

   ```
   wget https://github.com/mrirecon/bart/archive/v1.0.00.tar.gz
   tar xvfz v1.0.00.tar.gz
   cd bart-1.0.00
   ```

5. Build and install, or with GPU support as below:

   ```
   make
   sudo make install
   ```

6. Check that it works: at the Linux prompt `bart version`, and from a Windows command prompt `wsl bart version`.

Building BART with GPU support on Windows (provisional, highly recommended)

1. Install the NVIDIA driver in Windows, as described at https://docs.nvidia.com/cuda/wsl-user-guide/index.html. It is what gives WSL the GPU; no Linux display driver goes inside WSL.

2. Install the CUDA toolkit inside WSL, from https://developer.nvidia.com/cuda-downloads (Linux, x86_64, WSL-Ubuntu). For CUDA 12.8:

```
wget https://developer.download.nvidia.com/compute/cuda/12.8.0/local_installe
rs/cuda_12.8.0_570.86.10_linux.run
sudo sh cuda_12.8.0_570.86.10_linux.run
```

3. Set the environment variables, in the shell you build in: `CUDA_BASE` for the CUDA version installed, `GPUARCH_FLAGS` for the card. BART has to be built for the card's generation (look it up at https://developer.nvidia.com/cuda-gpus): a build for another one compiles without complaint and fails at the first calculation.

Generation	Cards	GPUARCH_FLAGS
Pascal	GTX 1060, 1080, Titan X	`-arch=sm_61`
Turing	RTX 2060 to 2080	`-arch=sm_75`
Ampere	RTX 3060 to 3090	`-arch=sm_86`
	A100	`-arch=sm_80`
Ada Lovelace	RTX 4060 to 4090	`-arch=sm_89`
Blackwell	RTX 5070 to 5090	`-arch=sm_120`

```
export CUDA=1
export CUDA_BASE=/usr/local/cuda-12.8
export CUDA_LIB=lib64
export GPUARCH_FLAGS="-arch=sm_61"
export PATH=$CUDA_BASE/bin:$PATH
export LD_LIBRARY_PATH=$CUDA_BASE/$CUDA_LIB:$LD_LIBRARY_PATH
```

4. Make. If the linking fails with an error about libstdc++ or libgcc, point the linker at the installed gcc with the last line (its number from `ls /usr/lib/gcc/x86_64-linux-gnu/`) and make again.

```
make -j$(nproc) CUDA=1 CUDA_BASE=$CUDA_BASE CUDA_LIB=$CUDA_LIB GPUARCH_FLAGS
="$GPUARCH_FLAGS"
sudo make install
export LDFLAGS="-L/usr/lib/gcc/x86_64-linux-gnu/13 $LDFLAGS"
```

5. Test: `make utest_gpu`

5. Running the software

Open **P2ROUD** from the Applications folder (macOS), or from the Start menu or the desktop shortcut (Windows). The P2ROUD logo appears while the app loads, and the window follows within seconds. At its start the app looks for BART, and the message box reports what it found; "Ready" says it is done. A scan given on the app's command line (an MRD file or a scan folder), as a program that starts the app with a scan does, is loaded at once, as **Load MRD** would load it. When the app is open already, a scan given this way goes to the open window instead: the window comes to the front and loads it, or loads it when the step under way ends.

Newer versions. At its start the app asks GitHub (https://github.com/Moby1971/MRI-apps) whether there is a newer release of it. The message box then shows the current and the newest version and, when there is a newer one, where to download it. Without a network connection it says that the version could not be checked, and the app works as usual. New versions also arrive by themselves while **auto update** in the Settings tab is ticked (see panel 5 below). With **auto update** unticked and **notify at start** ticked, a window then offers the newer version. `checkForUpdates` in the settings file switches the check off; `versionCheckTimeout` sets how many seconds it waits for GitHub (5).

This manual. The **Manual** button at the bottom of the window, with the app's icon on a book, opens this manual in the computer's PDF viewer.

The window has a fixed size. On a screen too small for it, the window fits the screen and can be scrolled.

The first Load MRD dialog opens in the newest exam of the scanner's MRD folder (on the scanner computer, Windows), or in the folder written in `mrdpath.txt`, a text file in the settings folder below; without either, in the folder of the scan you loaded last.

Settings and log files

P2ROUD keeps its files in a folder of its own, which it makes at its first start:

macOS	`~/Library/Application Support/P2ROUD/0.5.9p`
Windows	`%APPDATA%\P2ROUD\0.5.9p`

Settings. The settings are kept in `p2roudSettings.json`. The file is written with the standard values the first time a version starts, and can then be edited by hand; changes take effect at the next start. It holds, among others, the default compressed sensing weights and iterations for each type of data, whether the EPI corrections and the ZTE registration are ticked at start, the 3D UTE trajectory file chosen in the Radial tab, the spoke direction and frame settings of golden-angle radial scans (see *Golden-angle radial scans* in step 6), and the field strength written into DICOM headers made without the scanner's own when the scan's `.rpr` file gives none (7 T). A value that cannot be used is replaced by its standard value, and the message box says so.

P2ROUD keeps your choices of the theme, the sounds, the message level, **notify at start**, **auto update**, **CS reco**, **PCA**, **BMD**, **de-ring**, the FSE navigator and envelope, **Σ NE** and **Σ NR** and how they sum, the EPI corrections, the ZTE registration and dead time, and **auto** in the Images tab, and the folders last used for loading a scan and for an export, for the next start; a new version takes them over.

Log files. Every session writes a log file, `p2roud-log-<date and time>.txt`, to the same folder. The log keeps every message, also those the message box does not show at the chosen message level. The newest 50 log files are kept; older ones are removed when the app starts. The number is `logFilesToKeep` in `p2roudSettings.json`. No other file in the folder is touched.

Other files in the folder. `ktraj.txt`, the app's 3D UTE readout profile and the default trajectory file (see step 1), written at the start when it is not there; `3DUTEgradDelays.txt`, the six gradient delays a 3D UTE scan loads with (see *Reconstruction of radial scans*), written with zeros at the start when it is not there; `mrdpath.txt`, when you add it; and `graddelays.txt`, when you add one, which takes the place of the app's own gradient delays.

6. Basic operation



The P2ROUD program operates from a single window with 5 panels.

Panel 1: Dashboard. The task buttons and parameters that control the reconstruction process. The dashboard tasks need to be completed from top to bottom. A green light next to a task indicates that the task has been completed; yellow, that it is busy; red, that it has not been completed yet, or failed. At the bottom of the Dashboard one progress bar shows the progress of every task: what it is doing and how far it has got ("Loading", "Reconstruction", "Export 30 of 120", "Finished").

Panel 2: K-space / Coils / Metadata. The raw k-space data, and the coils and their sensitivity maps. The **Metadata** tab beside them holds the record of the scan: the animal, the session and the contrast agent (step 8).

Panel 3: Scan parameters / Geometry / Reconstruction / Radial. The acquisition, geometry, reconstruction and radial reconstruction parameters.

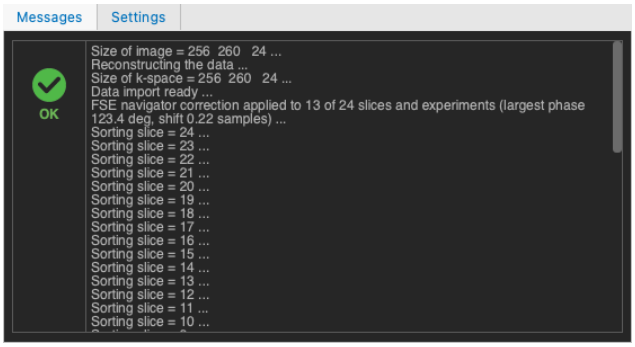
Panel 4: Images / MIP / Phase. After reconstruction, the reconstructed images.

Panel 5: Messages / Settings. Program status and messages, newest on top. In the Settings tab the sounds can be switched on or off (**sounds on/off**), the **light** or **dark** theme chosen (dark is the default), and the **message level** (1 to 3) set: how much the message box shows. **Updates**:

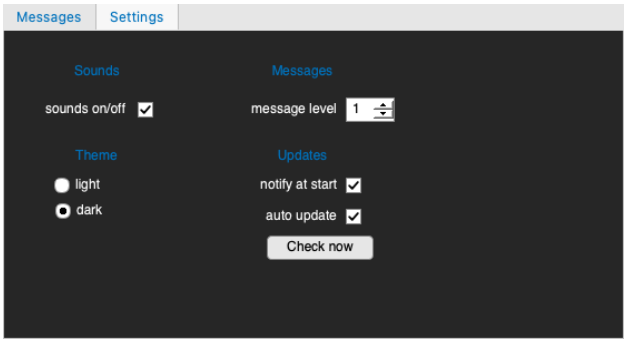
- **auto update**, ticked as it comes: new versions of P2ROUD arrive by themselves. When one is ready, a window says so with what is new: **Install and Restart** installs it and starts the new P2ROUD; **Later** installs it when you quit P2ROUD, without asking again. The window waits while the app is working, such as during a reconstruction or an export, or while a movie plays. Without an internet connection nothing is said and nothing waits. On Windows, P2ROUD installed for all users can be updated only by an administrator; for other users the message box says once where to get the new version to install by hand.
- **Check now** looks for a newer version at once, also with **auto update** off; the message box says what it found. On the Mac, **Check for Updates...** in the P2ROUD menu does the same.
- **notify at start**, ticked as it comes, counts only with **auto update** off: when a newer version is available, a window at the start offers it. **Download** gets the installer for this computer, **Skip this version** shows no window for that version again, **Later** asks again at the next start.

The message box says at every start whether P2ROUD is up to date. The status icon shows OK, WARNING, ERROR or BUSY.

Messages



Settings



Step 1: Loading data

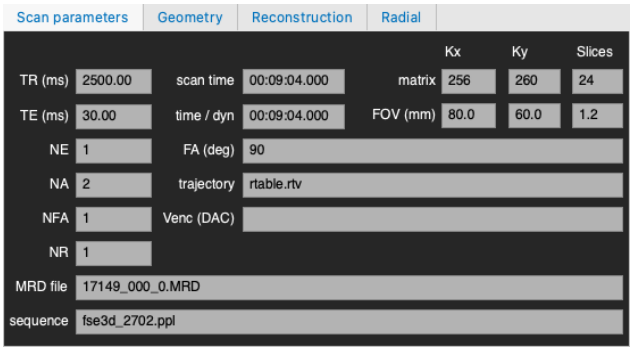
Press **Load MRD** to load the MRD raw data file.

P2ROUD can reconstruct spin-echo, fast spin-echo, gradient-echo, MPRAGE and CINE data, multi-slab 3D, 2D radial, 3D UTE and ZTE data, and 2D EPI (diffusion and fMRI). However, the program is particularly designed to reconstruct undersampled data with a user-defined undersampling pattern (e.g. SENSE, random undersampling, pseudo-spiral or radial).

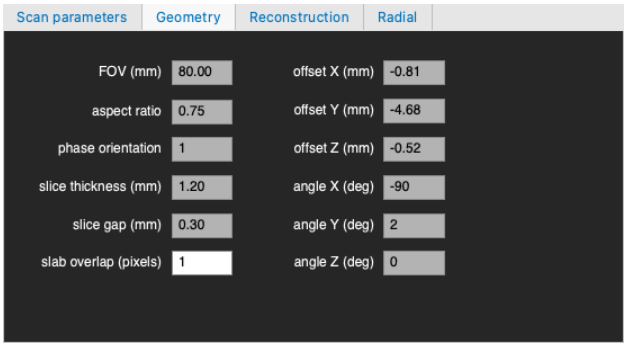
For data acquired with multiple receivers (phased-array RF coil) select one of the MRD files; the app finds the others. The relevant acquisition parameters are shown in panel 3.

A fast spin-echo scan is put in order with the scanner's k-space table, `rtable.rtv`, which the scanner keeps in the scan's `Recon_001` folder. When you copy a scan, copy that folder with it: without the table the message box warns that the images will be scrambled along the phase encoding.

Scan parameters



Geometry



k-space trajectory. There are three options to include a user-defined k-space trajectory in the FLASH (version 635e) sequence.

1. **2D data** — pe1-order = 3 ("LUT"). The k-space points for the 1st phase-encoding direction (views) are provided in the views table in the sequence. The trajectory field shows `mul_kspace_traj`.
2. **3D data** — pe1-order = 4 ("exLUT"). The k-space points for the 1st and 2nd phase-encoding directions (views and views2) are provided in a text file (`*.txt`), as many points as views x views2, as a list of points without comma separation, for example:

```
-48       (views    k-point 1)
0         (views2   k-point 1)
-47       (views    k-point 2)
0         (views2   k-point 2)
-46
1
...
```

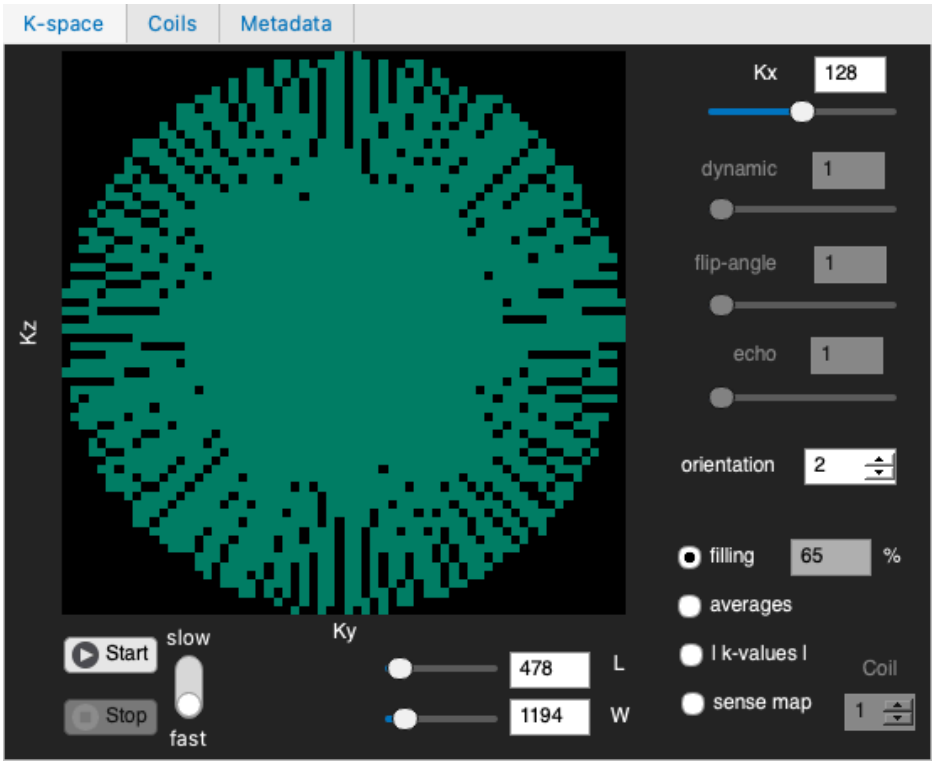
3. **Undersampled 2D and 3D data** — pe1-order = 5 ("nrLUT"). The file starts with the number of points, then the points as above, fewer than views x views2.

The trajectory file (options 2 and 3) may have any name; keep it beside the MRD file. The message box names the file used, or, when none fits the scan, says why. (3D UTE scans read theirs from `lut*.txt`.)

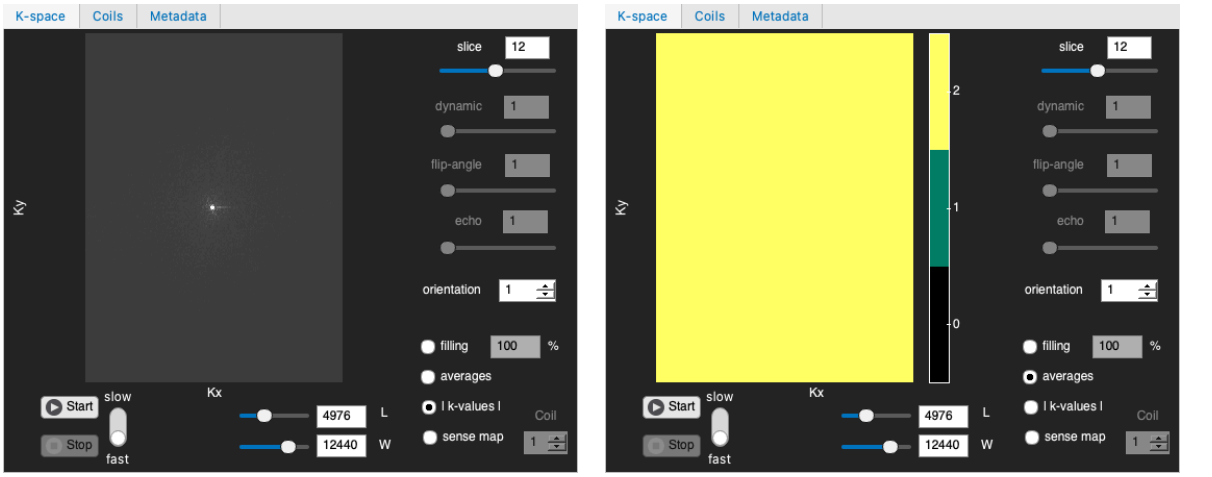
The readout profile of 3D UTE data is read from the trajectory file chosen in the UTE section of the Radial tab (step 6), by default `ktraj.txt` in the settings folder. The file has one column for all axes or three for x, y and z. A comment line in it gives the sample period the profile was made for (`% sample period 2.5 us`; a file without one is taken as 2.5 µs). A comment line can also give the data offset (for example `% data offset 7.5`), which is then used instead of the one in the Radial tab, for scans with the file's sample period.

Step 2: Inspecting k-space

The raw k-space data is displayed in the K-space tab of panel 2. The radio buttons show the acquired k-space points (**filling**), the averages per k-point (**averages**), the absolute value of k-space (**| k-values |**) or a coil sensitivity map (**sense map**). The percentage of k-space points acquired is shown beside **filling** (65 % for the dataset below).



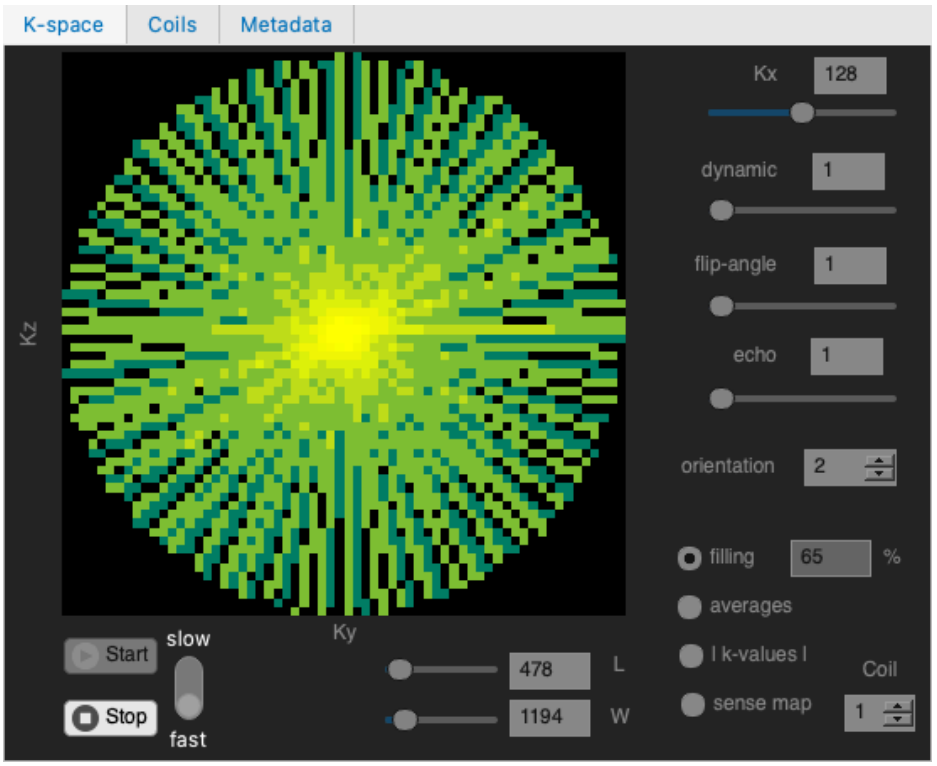
| k-values | averages



The sliders step through the third k-space dimension (**Kz**, or **slice** for 2D data), the repetitions (**dynamic**), the flip angles (**flip-angle**) and the echoes (**echo**), if applicable; for a series that steps through TR, TI or CINE frames the slider is named so. The **orientation** spinner chooses the k-space plane shown, the **Coil** spinner the coil, and **L** and **W** set the level and window.

Step 3: The k-space trajectory

Press **Start** to start an animation of the k-space trajectory, and **Stop** to stop it. The switch beside them toggles between **fast** and **slow** animation.



Step 4: Fourier transform reconstruction

For a first quick inspection of the reconstructed images, press **Reco** with **CS reco** unticked: a Fourier transform reconstruction.

X, **Y** and **Z** in the Dimensions panel set the size of the images: for Cartesian data k-space is zero-filled or cropped, for radial, UTE and ZTE data the images are resized. For 2D data the third field is **Slices** and keeps the number of slices acquired. **NR** sets the number of repetitions (see step 6).

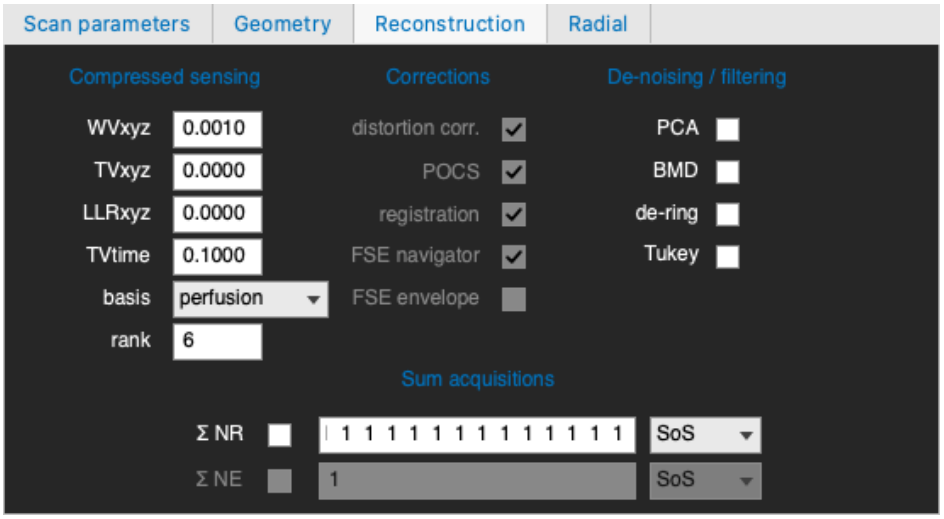
Step 5: Coils

For acquisitions with a phased-array coil the Coils tab gives access to the channels: coils can be checked or unchecked to use or omit them in the reconstruction. Relative sensitivities can be set by hand, or determined with **auto sensitivity**, which for a CS reconstruction also computes the coil maps shown under **Sensitivity maps**. Auto sensitivity is ticked at every load of multi-coil data.



Step 6: Reconstruction

Ticking **CS reco** in the Dashboard gives access to the Compressed Sensing (CS) reconstruction parameters in the Reconstruction tab of panel 3.



The CS regularization parameters are:

1. **WVxyz** — wavelet regularization in the spatial (x, y and z) dimensions.
2. **TVxyz** — total variation constraint in the spatial dimensions.
3. **LLRxyz** — locally low-rank constraint in the spatial dimensions.
4. **TVtime** — total variation constraint in the dynamics (repetitions, NR) dimension. It is on by default (0.1) whenever more than one dynamic or frame is reconstructed, and 0 for a single image; 3D Cartesian scans keep 0.05 for both. The message box says when the app sets it. TV over time is not used across diffusion volumes of different b-values or directions: there the field is greyed. You can still type 0 for a time series: your value stays until the scan becomes a single image or is loaded again.

The spatial dimensions of the reconstructed data can be increased or decreased. The number of repetitions (**NR**) can also be changed to undersample or average data in the dynamics dimension.

Frames shorter than the dynamics. For a 2D multi-dynamic scan acquired along a list of k-space lines (a perfusion scan, for example), **NR** can be set higher than the number of dynamics acquired: each dynamic is then split into shorter frames, each made of fewer k-space lines. Example: 50 dynamics with **NR** 400 give 8 frames per dynamic. When **NR** goes above the dynamics acquired, **CS reco** is ticked, which suits such frames, and **TVtime** is on as for any time series; the message line says so. When **NR** returns to the dynamics, **CS reco** goes back to what it was, unless you changed it yourself in the meantime.

basis and **rank** (2D multi-dynamic scans with CS reco) describe how the signal of a pixel may change from frame to frame:

- **none** — every frame is reconstructed with the regularization above (the default).
- **perfusion** — the signal follows a few curves of a perfusion model, made from the scan itself, besides a constant. The curves are smooth, which gives frames with little noise. The model takes the time of a frame from the scan's duration.
- **data-derived** — the signal follows the time courses found in the centre of k-space over the frames.

rank is the number of curves. It is a proposal: after every sort (a load, a new **NR**), and when the basis is chosen, the app puts a rank in the field and says in the message box why ("Temporal basis perfusion: rank 7 proposed for 400 frames ..."). For **perfusion** it counts the model curves at least 2 % as strong as the strongest; for **data-derived** it proposes 10 (at most the frames); with **TVtime** on, at most 12. You can type another rank: the reconstruction uses the number in the field. A rank you typed stays until a new proposal is due (a new sort, the basis chosen again, or **TVtime** switched on or off); the message box then says that it is replaced. The basis you chose is kept for the next start, the rank is not.

A higher **rank** follows faster or more varied changes and keeps more noise; a lower one gives smoother curves. With **TVtime** or **LLRxyz** above 0 the basis is reconstructed by the app's own engine, which applies them along the frames; without them by BART where it is installed. The basis is off for other scans.

Pressing **Reco** starts the CS reconstruction. While a reconstruction runs, the **Reco** button reads **Stop**; pressing it stops the reconstruction and leaves the earlier images and the parameters as they were.

For multi-slab 3D (MOTSA) scans, the slabs are put where they were acquired and blended over **slab overlap (pixels)**, in the Geometry tab, on either side of each junction.

Further options in the Reconstruction tab

De-noising / filtering is applied to the reconstructed images; the methods can be combined.

- **PCA** removes noise by principal component analysis over the images of a series (dynamics, diffusion directions, flip angles, echoes).
- **BMD** applies block-matching denoising.
- **de-ring** suppresses Gibbs ringing. It is not available for radial, UTE and ZTE data.

Corrections.

- **Tukey** tapers the edges of k-space against ringing, at some cost in resolution; loading a Cartesian scan unticks it, a radial or UTE scan ticks it.
- **FSE navigator** (ticked by default) corrects the echoes of a fast spin echo scan with its navigator.
- **FSE envelope** reduces the ghosting of a fast spin echo scan at the same signal-to-noise ratio; the images are slightly less sharp.

The FSE boxes are available for fast spin echo scans with a navigator.

Sum acquisitions. With **Σ NR** or **Σ NE** ticked, the dynamics or the echoes are combined into one image by **SoS**, **Max** or **Sum**; the field beside the box selects which, with a 1 or 0 for each.

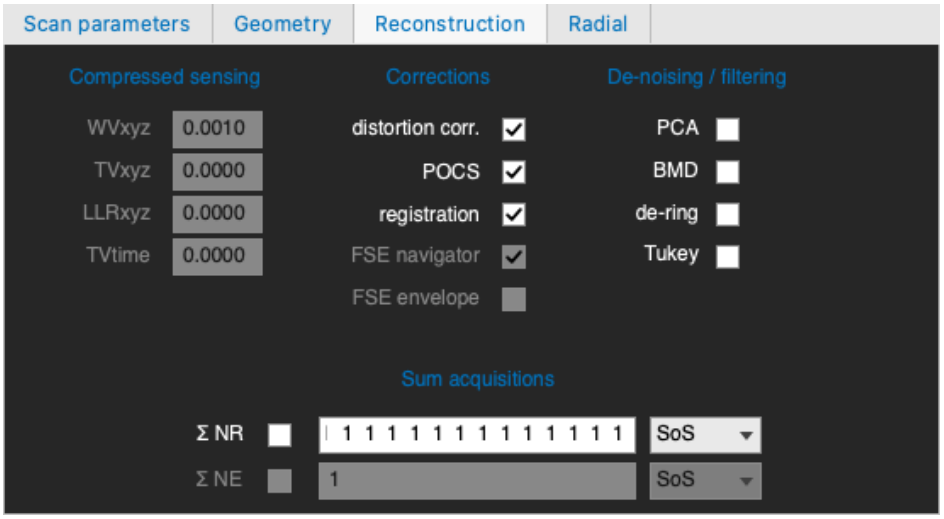
EPI scans

2D EPI scans, single-shot or read-interleaved, spin echo or gradient echo, with or without partial Fourier, are reconstructed as diffusion series (DTI, several b-values) or fMRI time series. The images are masked when the study has a mask scan of the same geometry (named in the scan settings, or the nearest one). Three corrections in the EPI part of the tab are on by default:

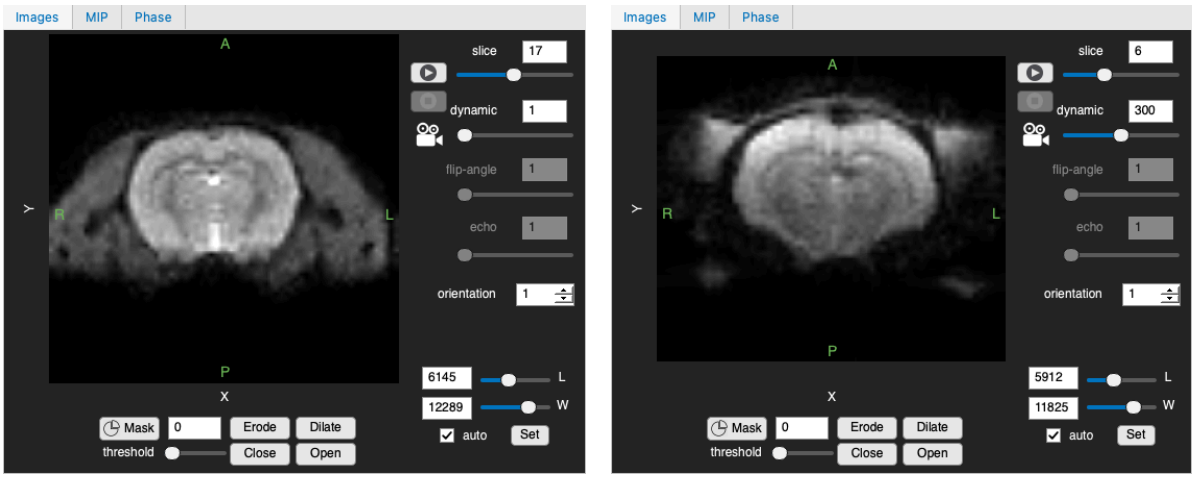
- **distortion corr.** corrects the geometric distortion with a partner scan of the same study, with the same slices and echo train and the opposite phase-encoding direction, which the app finds by

itself. Without a partner scan the message box says so and nothing is changed.

- **POCS** completes a partial Fourier scan: sharper images, at some cost in signal-to-noise ratio.
- **registration** aligns the volumes of the series in the image plane.



Diffusion (DTI, 130 volumes): a b = 0 volume **fMRI (600 dynamics): slice 6, dynamic 300**

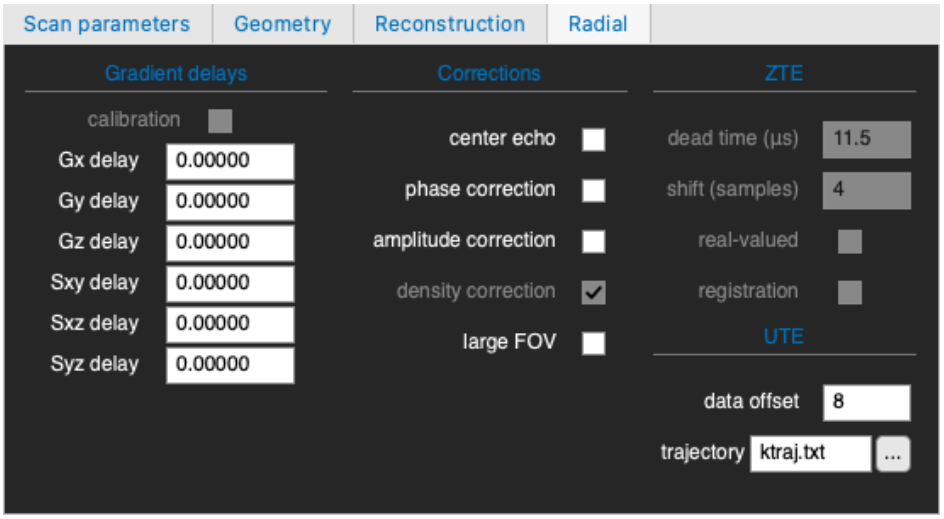


The **DCM / NIFTI** export of an EPI series also writes a BIDS dataset in `BIDS/sub-<study>/ses-1`: `func` (fMRI) or `dwi` with the gradient table (diffusion), and `anat`, the fast spin echo scans of the study (T1w and/or T2w), which the app finds and reconstructs itself. An fMRI series needs such a scan, a diffusion series does not.

For a diffusion scan with at least 6 directions and a b = 0 volume, MRtrix3 and DSI Studio scripts for tractography (`tractography_MRtrix_<tag>` and `tractography_DSIStudio_<tag>`, `.sh` and `.bat`) are written into the NIfTI folder; the app does not run them.

Reconstruction of radial scans

The Radial tab gives access to advanced reconstruction parameters for 2D radial, 3D UTE and ZTE scans.

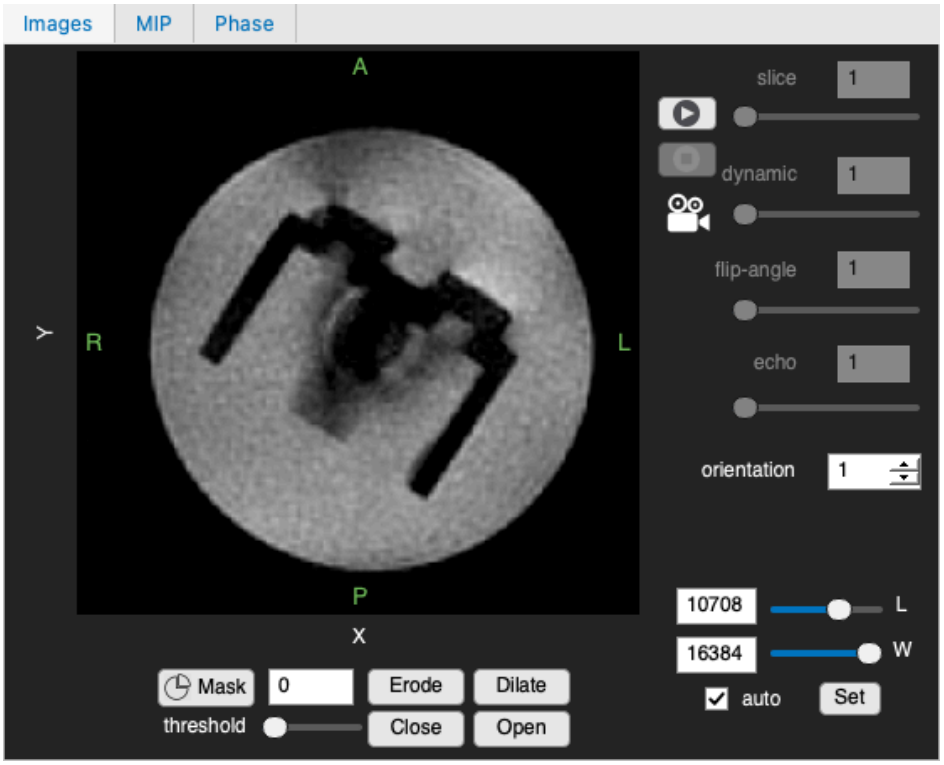


Tick **calibration** to estimate the gradient delays of a 2D radial scan from the data at the next reconstruction; the delay fields are then filled in and the checkbox is unticked. 2D radial scans load with calibration ticked; the calibration takes a few seconds, and **center echo** does not apply. **phase correction** corrects the phase of every spoke. For 3D UTE and ZTE scans the delays are entered in the fields. The delays, in samples, are **Gx**, **Gy** and **Sxy** for 2D radial data, and **Gx**, **Gy**, **Gz**, **Sxy**, **Sxz** and **Syz** for UTE and ZTE; they are reset when a data set is loaded. A 3D UTE scan loads with the six delays of `3DUTEgradDelays.txt` in the settings folder: six numbers in that order, all 0 as the app writes the file. Put in your own values to have every 3D UTE scan start with them; lines starting with `%` are comments. A file that cannot be read, or does not hold six numbers, is replaced by a new one with zeros, and the message box says so.

The **UTE** section applies to 3D UTE scans only; for other scans it is grayed out. **data offset** is the number of samples at the start of each readout before the trajectory starts; these are left out. It may have decimals: with 7.5 the result lies between those of 7 and 8. **trajectory** names the file with the readout profile, `ktraj.txt` by default. Type a name for a file in the settings folder, or click **...** to choose a file; a file outside the settings folder is kept with its full path. The choice is kept in the settings file for the next start, and a loaded 3D UTE scan uses the new file at once. When the file is missing or cannot be used, the message box says so and the app's own profile is used.

2D radial images are black outside the circle the spokes cover, in the corners of the image.

The Images tab below shows a reconstructed 2D radial scan of a phantom.



Golden-angle radial scans

P2ROUD reconstructs the golden-angle radial scans of the aumc_flash_004 and aumc_flash_005 sequences, as 2D slices or as a stack of stars (a 3D slab measured as radial spokes in every partition). With phase-encoding order 1 the angle of the spokes steps by the golden angle, with order 2 by the tiny golden angle, and it keeps stepping from one repetition to the next, so that any run of consecutive spokes covers k-space evenly. At the load the message box names the sequence, the angle step and the number of spokes per repetition. Order 0 (uniform radial) in 2D is reconstructed as any 2D radial scan. Scans with order 3 to 5, with more than one average or echo, uniform radial as a stack of stars, or a stack of stars with several slabs are not reconstructed, and the message box says why; for more spokes, measure repetitions instead of averages.

Frames. The images form a series of frames. **NR** sets the number of frames: the spokes of all repetitions, in the order measured, are divided over them. **NR** may be higher than the number of repetitions, which gives shorter frames of fewer spokes. Example: 4 repetitions of 48 spokes with **NR** 6 give 6 frames of 32 spokes. By default the frames follow one another without overlap. `goldenAngleSpokesPerFrame`, in the `reconstruction` block of the settings file, gives every frame that many spokes instead (0, the default, divides the spokes as above). The frames then start evenly spread over the series, and when a frame is longer than the step between two frames, neighboring frames share spokes: a sliding window. Example: 192 spokes with **NR** 5 and 64 spokes per frame give 5 frames of 64 spokes, one starting every 32 spokes. After each reconstruction the message box gives the number of frames, their spokes, the step between them and the time a frame covers.

calibration and **phase correction** work as for other 2D radial scans, and a golden-angle scan loads with both ticked.

Spoke direction. Two settings in the `site` block of the settings file match the spokes to the image axes: `goldenAngleSign` (1, or −1 for the opposite sense of rotation) and `goldenAngleOffset` (degrees, 0). Set them once, from the first golden-angle scan of a phantom whose shape shows left from right

and top from bottom, against a Cartesian scan of the same slice:

1. When the golden-angle image is mirrored against the Cartesian one, set `goldenAngleSign` to −1.
2. When it is turned, set `goldenAngleOffset` to the angle in degrees that brings it onto the Cartesian image; when the image turns the other way, use the opposite sign.

Changes in the settings file take effect at the next start of the app. The values then hold for every golden-angle scan.

Center-out spokes (aumc_flash_005). With the read prephaser off and order 1 or 2, aumc_flash_005 measures center-out half spokes, for a short echo time: every spoke starts at the center of k-space, and the angles cover the full circle. The message box says so at the load. The images have the matrix prescribed on the scanner; the message box says how many samples of each spoke are used. **calibration** does not apply to these spokes (the message box says so when it is ticked): enter the delays in **Gx**, **Gy** and **Sxy** instead. **phase correction** corrects the phase of every spoke. With the read prephaser on, aumc_flash_005 is reconstructed as aumc_flash_004.

`centerOutRampOffset`, in the `site` block of the settings file (µs, 0 by default), sets where the samples of a center-out spoke lie in k-space. A wrong value blurs the images. Set it from the first center-out scan of a phantom with sharp edges, such as a tube: reconstruct the scan with a few values, for example from −10 to 10 µs in steps of 2 µs (each change takes effect at the next start of the app), and keep the value that gives the sharpest edges and the most even signal inside the phantom. The value holds for one receiver bandwidth (sample period): set it again for scans with another bandwidth.

3D UTE scans

For 3D UTE, **center echo** centers the echoes before reconstruction and **phase correction** corrects their phase; a 3D UTE scan loads with both unticked. It loads with the **data offset** that fits the scan's protocol, which P2ROUD works out from the scan's own parameters (its echo time and sample period among them), and the message box gives it at the load. A ute3d22 scan at a sample period of 2.5 µs loads with 0. A scan whose parameters give no offset loads with 6 (sample period 2.5 µs) or 0.6 (5 µs). An offset you type in the field, or one named in the trajectory file, is used instead.

P2ROUD reconstructs the 3D UTE scans of the ute3d22, ute3d23 and aumc_ute3d_001 sequences. Their volumes are shown and exported in the scanner's orientation: the read, phase and slice directions are those of a Cartesian scan of the same object, and an off-center scan lies where it was measured. The message box says when a volume is moved to its off-center place.

large FOV lets less signal leak into a 3D UTE image from beyond the field of view; it takes about a quarter longer. It applies to the NUFFT reconstruction, not to CS.

density correction performs a 2D or 3D density correction in k-space to account for non-uniform sampling. It only applies to CS reconstructions, and its checkbox is on only with **CS** ticked.

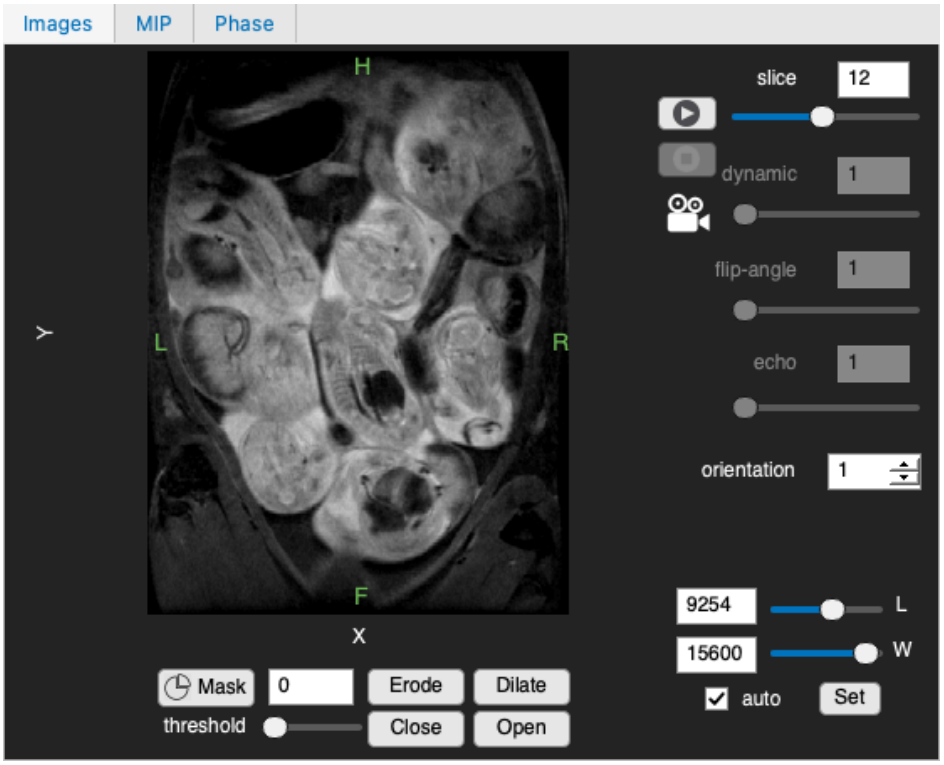
amplitude correction evens out the signal amplitude of the spokes. The Tukey filter, which tapers the readout, is set in the Reconstruction tab.

ZTE scans have no calibration and none of these corrections. Their **dead time (µs)** is the delay between the excitation and the first sample, 11.5 µs by default, and **shift (samples)** the first sample kept, 4 by default. Too small a dead time leaves solid objects dark inside with bright edges, too large a value makes them bulge in the middle. When the first sample of the spokes still holds leftover RF pulse

signal, P2ROUD leaves that sample out and says so in the messages; the dead time field keeps its value. **real-valued** gives a darker background while the object keeps its signal. **registration** aligns the dynamics of a series in 3D.

Step 7: Viewing the reconstructed images

After reconstruction the MR images are displayed in panel 4.



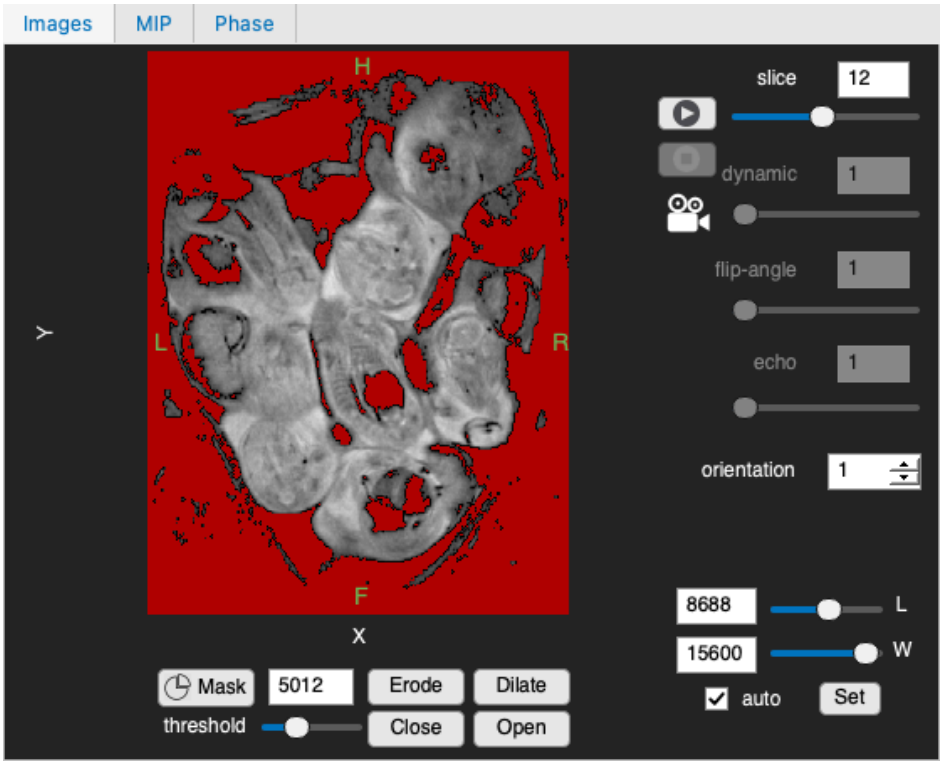
The image orientation can be changed with the **orientation** spinner; the green letters around the image give the anatomical directions. The MIP and Phase tabs follow it. On an oblique plane an edge can carry two letters, the main direction first: **LA**, for example, points mostly left and partly anterior.

Image intensity can be adjusted with the level (**L**) and window (**W**) sliders. Checking **auto** adjusts the intensity automatically; **Set** takes the level and window of the image shown and unticks auto.

Press the play button to start a movie of the images as a function of the dimension shown next to the camera icon (for example the dynamics), and the stop button to stop it.

When the pointer is over the image, a small toolbar appears: its zoom button zooms with a box, a click or the wheel, and its home button (**Whole image**) shows the whole image again. With zoom off, the scroll wheel steps through the slices. The scroll wheel over a slider moves that slider, one step per notch.

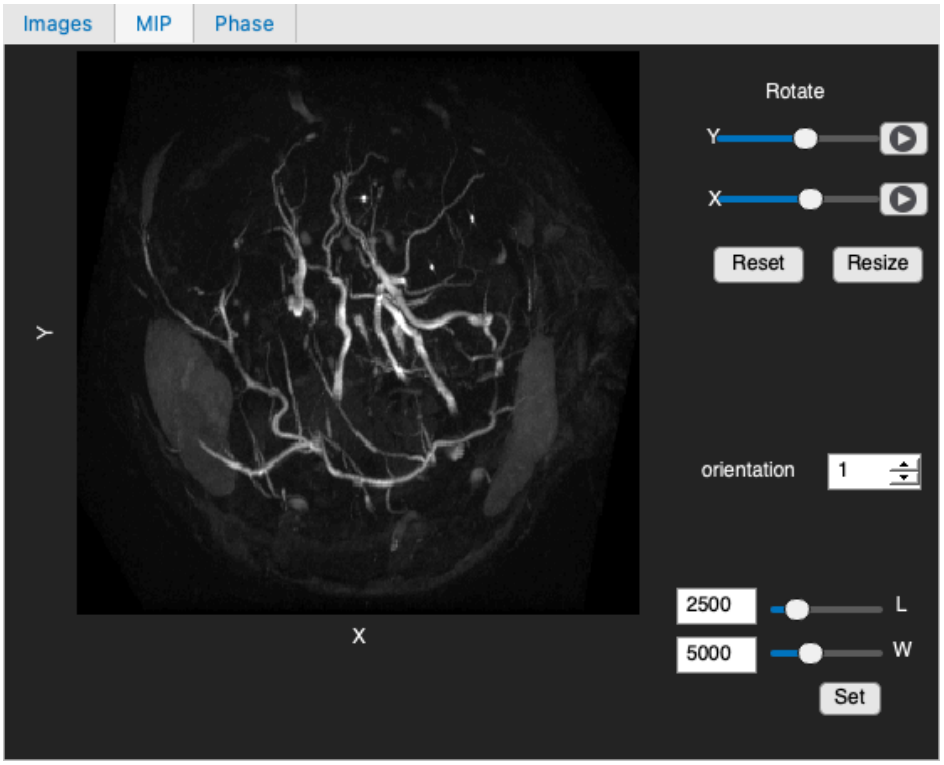
Images can be segmented (on the basis of a threshold) by pressing **Mask**. Use the **threshold** slider or the morphological operations (**Erode**, **Dilate**, **Open**, **Close**) to fine-tune the segmentation. The masked-out pixels are shown in red. The mask is used for display, the MIP and phase unwrapping only.



Maximum intensity projection (MIP)

In the MIP tab a maximum intensity projection is shown. The MIP can be turned with the **X** and **Y** sliders under **Rotate**, and animated with the play buttons beside them, 5° per step (pressing again stops). **Reset** stops the movies and returns to the start; **Resize** halves the volume for speed. The MIP has its own level, window and **Set**.

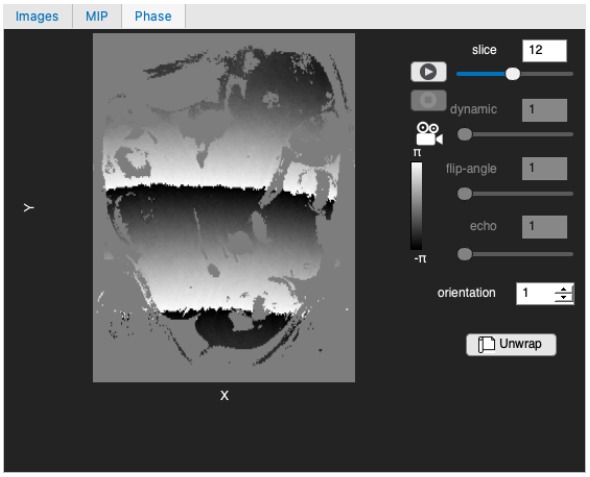
The object keeps its size as it turns: the view is the same at every angle, and the corners of a turned projection that reach beyond it are cut off.



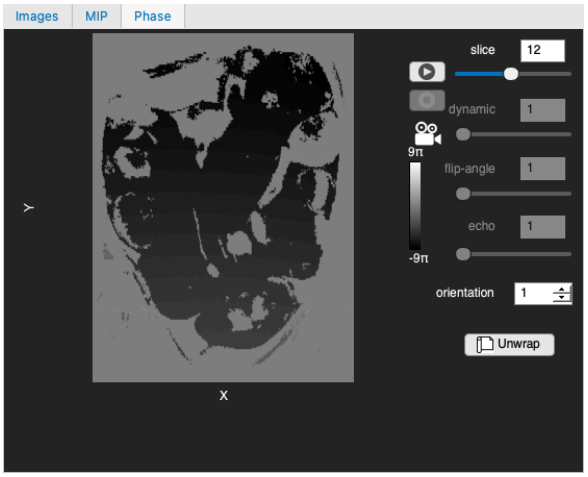
Phase images

The Phase tab shows the phase images, with their own sliders, movie and orientation. The phase can be unwrapped by pressing the **Unwrap** button; pixels outside the mask keep their wrapped value.

Phase



Unwrapped

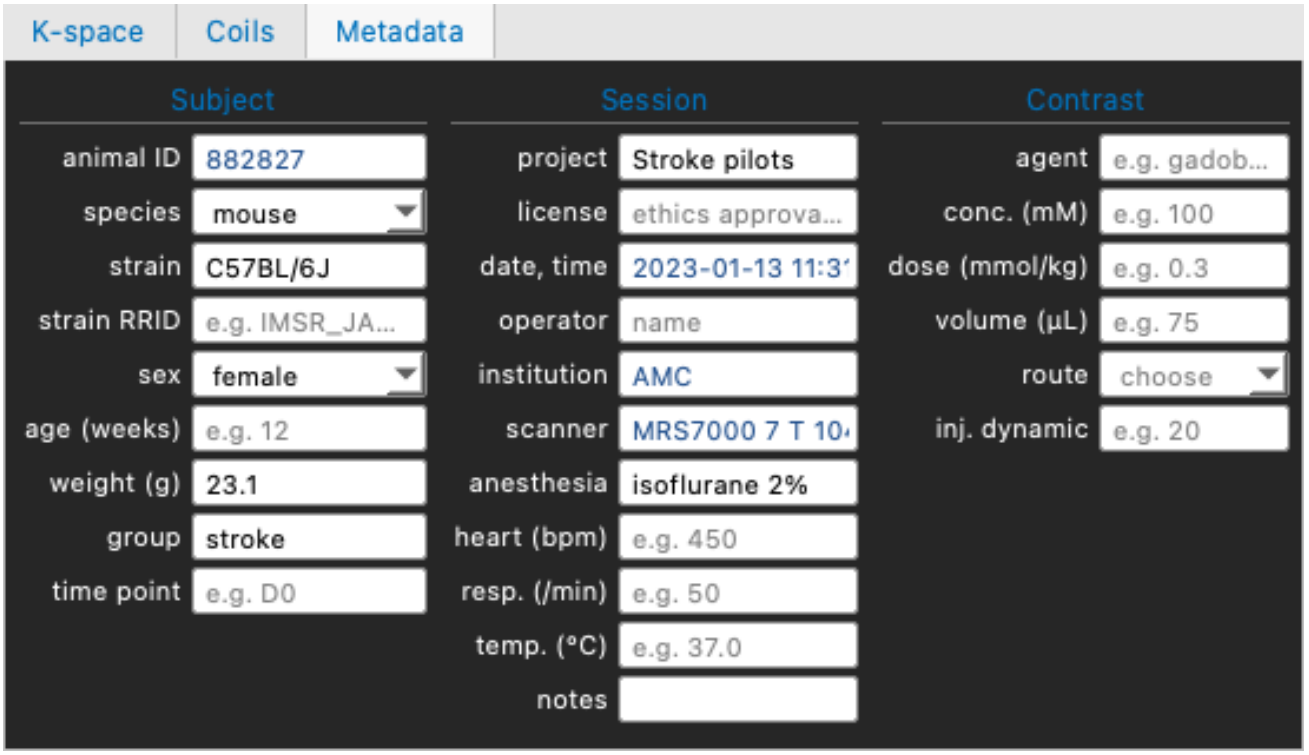


Step 8: Metadata

The **Metadata** tab, beside the K-space and Coils tabs, holds the record of the scan, which the DCM / NIFTI and GIF exports write with the images. Its fields stand in three columns:

- **Subject:** animal ID, species, strain and its RRID, sex, age, weight, group and time point;
- **Session:** project, ethics license, date and time, operator, institution, scanner, anesthesia, heart and respiration rate, and temperature, with a field for notes;

- **Contrast:** agent, concentration, dose, volume, route and the dynamic it was injected at.



At a load the program fills in what the scan's files say: the animal ID and the time point (from the patient name typed at the console: `20240212_926752_D0` is animal 926752, day 0), the date and time, the institution and the scanner, and the project. What the console leaves at its standard values (sex O, weight 0, the login as the operator) is left empty.

What was filled in for an earlier scan of the same exam is filled in again: the animal's details are typed only once per exam. The date, the heart and respiration rate, the temperature and the notes belong to one scan and are never taken from another.

Your institution, operator, anesthesia, species, strain and strain RRID are kept for later exams: they are shown, in blue, from the start, and a scan's own value (from its files or its exam) takes their place where it has one.

A value the program filled in is shown in blue; every field can be corrected, and then counts as yours. Species, sex and route are lists to choose from, or to type in; numbers are kept with their units (a number field holding something else is edged in red). The species and sex are written with their codes (NCBI Taxonomy, PATO), so the record can be read by other software as well as by people.

Step 9: Image export

There are three ways to export the images, in the Export panel of the Dashboard.

1. **DCM / NIFTI** — exports the images as DICOM files for further processing in third-party software. The DICOM files carry the species, strain, sex, age and weight of the animal where the Metadata tab has them. The images are also written as NIfTI files with their geometry, and for diffusion scans with `.bval` and `.bvec` files; `recoparameters_<tag>.txt` records the scan and the settings used. For the scanner's spin-echo diffusion EPI sequences (the current ones, their older versions from 2020 on, and the lab's corrected version, with an EPI readout or one echo per shot) every

volume's b-value and direction come from all the gradients the sequence plays, the imaging gradients included. The volumes without diffusion weighting get their few s/mm² and a direction too, and `<image>_bmatrix.txt` beside the `.bval` and `.bvec` holds each volume's full b-matrix (Bxx Bxy Bxz Byy Byz Bzz in s/mm², in the axes of the `.bvec`). The messages say which b-values were used, and warn when the scan's protocol computed its b-values with another gradient strength or when the sequence reads the diffusion-weighted volumes at another echo time than the volumes without diffusion weighting (with the echo time that avoids it). For EPI series a BIDS dataset and scripts for tractography are written as well (see "EPI scans"). When the scanner's DICOM export of the scan is found in `…/Data/<study>/DICOM/<scan>`, the files go into `…/Data/<study>/DICOM/<series>P/<n>` and `Nifti/<tag>P/<n>`, a new numbered folder `<n>` each time; otherwise into a folder that is asked for (`DICOM/<tag>P/<n>` there). In the names of the folders and files the app makes, `<tag>` is the scan's name with every character other than letters, digits, `-`, `_` and `.` (a space, for example) written as `_`.

2. **GIF** — the data are exported as (animated) GIF files, one per slice, flip angle or echo. A zoomed image is exported as far as it is in view. When the MRD file lies in `…/MRD/<study>/<scan>`, they go into `…/Data/<study>/GIF/<tag>P/<n>`, a new numbered folder each time; otherwise into a folder that is asked for.
3. **MRD** — the data are stored as a new MRD file, `p2roud_<n>_<tag>.MRD`, in the folder of the input MRD file (or in a folder asked for, when that cannot be written), with `recoparameters_<tag>.txt` and, if the program finds an input `.rpr` file, a new `.rpr` file. The scan's `rtable.rtv` is renamed `origrtable.rtv.orig`. Additionally, when the program runs on the scanner computer (Windows), the data are reconstructed by the scanner software and new SUR files, and for nrLUT scans DICOM images made from them, are placed in the data directory for viewing and planning of next scans in the MR Solutions preclinical software. nrLUT scans are exported to MRD by themselves after every reconstruction.

Where an export asks for a folder, the dialog opens in the folder you chose last for an export.

The DCM / NIFTI and GIF exports also write, in `…/Data/<study>/Metadata/<tag>P/<n>` (or under the folder asked for):

- `metadata.json` — the Metadata tab's record;
- `p2roud-report_<tag>.json` and `.txt` — the report of the export: the program and its libraries, the scan and a checksum of each of its files, the steps of the session, the scan and reconstruction settings, and every file the export wrote with its checksum;
- `ro-crate-metadata.json` — the same as an RO-Crate (https://www.researchobject.org/ro-crate/), a standard description of the folder for data repositories.

The paths in them are relative to the study folder, so they hold no names of your own folders.

Step 10: Exit

Press **Exit** to shut down the program. Closing the window does the same, also while the program is busy: the window disappears, and the program closes as soon as the running step has finished.

7. Credits

- **BART toolbox and compressed sensing algorithms** — https://mrirecon.github.io/bart/. Michael Lustig, David Donoho, John M. Pauly, Sparse MRI: The application of compressed sensing for rapid MR imaging, Magn Reson Med 58, 1182–1195 (2007).
- **Data-derived temporal basis** — Han et al., J Cardiovasc Magn Reson 23, 8 (2021).
- **sigpy** — Frank Ong and Michael Lustig, SigPy: a python package for high performance iterative reconstruction, Proc. Intl. Soc. Mag. Reson. Med. 27, 4819 (2019). Used for compressed sensing without BART.
- **Gibbs ringing suppression** — Kellner, E., Dhital, B., Kiselev, V.G. and Reisert, M., Gibbs-ringing artifact removal based on local subvoxel-shifts. Magn Reson Med 76(5), 1574–1581 (2016).
- **NUFFT** — https://web.eecs.umich.edu/~fessler/
- **Phase unwrapping** — M. A. Herráez, D. R. Burton, M. J. Lalor, and M. A. Gdeisat, "Fast two-dimensional phase-unwrapping algorithm based on sorting by reliability following a noncontinuous path", Applied Optics, Vol. 41, Issue 35, pp. 7437–7444 (2002); H. Abdul-Rahman, M. Gdeisat, D. Burton, M. Lalor, "Fast three-dimensional phase-unwrapping algorithm based on sorting by reliability following a non-continuous path", Proc. SPIE 5856, Optical Measurement Systems for Industrial Inspection IV, 32 (2005).
- **PCA denoising** — J. Veraart, D. S. Novikov, D. Christiaens, B. Ades-Aron, J. Sijbers, E. Fieremans, Denoising of diffusion MRI using random matrix theory, NeuroImage 142, 394–406 (2016); L. Cordero-Grande, D. Christiaens, J. Hutter, A. N. Price, J. V. Hajnal, Complex diffusion-weighted image estimation via matrix recovery under general noise models, NeuroImage 200, 391–404 (2019).
- **Block-matching denoising**
  - Y. Mäkinen, L. Azzari, A. Foi, "Collaborative Filtering of Correlated Noise: Exact Transform-Domain Variance for Improved Shrinkage and Patch Matching", IEEE Transactions on Image Processing, vol. 29, pp. 8339–8354 (2020).
  - M. Maggioni, V. Katkovnik, K. Egiazarian, A. Foi, "Nonlocal Transform-Domain Filter for Volumetric Data Denoising and Reconstruction", IEEE Transactions on Image Processing, vol. 22, pp. 119–133 (2013).
  - Y. Mäkinen, S. Marchesini, A. Foi, "Ring Artifact and Poisson Noise Attenuation via Volumetric Multiscale Nonlocal Collaborative Filtering of Spatially Correlated Noise", Journal of Synchrotron Radiation, vol. 29, pp. 829–842 (2022).
- **NIfTI conversion** — dcm2niix: C. Rorden et al., https://github.com/rordenlab/dcm2niix.